

4-2



Basis

Why nonzero?

- Let V be a nonzero subspace of $\mathcal{R}^n$. A basis B for V is a linearly independent generation set of V .

2. $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis for $\mathcal{R}^n$.

1. $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is independent

2. $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ generates $\mathcal{R}^n$.

$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathcal{R}^2$

$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$ $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ any two independent vectors form a basis for $\mathcal{R}^2$

Is it a Basis?

S subspace V
basis?

Assume that $\dim V = k$. Suppose S is a subset of V with k vectors

$S \subset V$

If S is independent



S is basis

By the extension theorem, we can add more vector into S to form a basis.

However, S already have k vectors, so it is already a basis.

If S is a generation set



S is basis

By the reduction theorem, we can remove some vector from S to form a basis.

However, S already have k vectors, so it is already a basis.

73 Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for a k -dimensional subspace V of $\mathbb{R}^n$. Prove that $\{\mathbf{v}, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is also a basis for V , where $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_k$.

$\dim V = k$ $\mathbf{v}, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$ k vectors

$\{\mathbf{v}, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ ind.?

$$c_1 \mathbf{v} + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow c_1 (\mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_k) + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow c_1 \mathbf{u}_1 + (c_1 + c_2) \mathbf{u}_2 + \dots + (c_1 + c_k) \mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow \underbrace{c_1}_0 = \underbrace{c_1 + c_2}_0 = \dots = \underbrace{c_1 + c_k}_0 = 0 \Rightarrow c_1 = c_2 = \dots = c_k = 0$$

73 Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for a k -dimensional subspace V of R^n . Prove that $\{\mathbf{v}, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is also a basis for V , where $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_k$.

BCV, B: k vectors?

B ind.?

74 Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for a k -dimensional subspace V of R^n , and let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, where

$$\mathbf{v}_i = \mathbf{u}_i + \mathbf{u}_{i+1} + \dots + \mathbf{u}_k \text{ for } i = 1, 2, \dots, k$$

Prove that B is also a basis for V .

76 Let $T: R^n \rightarrow R^m$ be a one-to-one linear transformation and V be a subspace of R^n . Recall that $W = \{T(\mathbf{u}) \in R^m: \mathbf{u} \text{ is in } V\}$ is a subspace of R^m .

(a) Prove that if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a basis for V , $\{T(\mathbf{u}_1), T(\mathbf{u}_2), \dots, T(\mathbf{u}_k)\}$ is a basis for W . $\dim W = ?$ $\rightarrow$ Ind. $\&$ generation see

(b) Prove that $\dim V = \dim W = k$

$$T(\mathbf{u}) = A\mathbf{u}$$

① Ind. ?

$$c_1 T(\mathbf{u}_1) + c_2 T(\mathbf{u}_2) + \dots + c_k T(\mathbf{u}_k) = \mathbf{0}$$

$$\Rightarrow c_1 A\mathbf{u}_1 + c_2 A\mathbf{u}_2 + \dots + c_k A\mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow A(c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k) = \mathbf{0}$$

$$\Rightarrow c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow c_1 = c_2 = \dots = c_k = 0$$



76 Let $T: R^n \rightarrow R^m$ be a one-to-one linear transformation and V be a subspace of R^n . Recall that $W = \{T(\mathbf{u}) \in R^m: \mathbf{u} \text{ is in } V\}$ is a subspace of R^m .

(a) Prove that if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a basis for V , $\{T(\mathbf{u}_1), T(\mathbf{u}_2), \dots, T(\mathbf{u}_k)\}$ is a basis for W . (incl. ✓)

(b) Prove that $\dim V = \dim W$.

② gen? $\underline{x} \in W$

$$x = T(u), u \in V$$

$$u = c_1 u_1 + c_2 u_2 + \dots + c_k u_k$$

$$x = T(c_1 u_1 + c_2 u_2 + \dots + c_k u_k)$$

$$\Rightarrow x = T(c_1 u_1) + T(c_2 u_2) + \dots + T(c_k u_k)$$

$$\Rightarrow x = \underline{c_1} T(u_1) + \underline{c_2} T(u_2) + \dots + \underline{c_k} T(u_k)$$

77 Let V and W be nonzero subspaces of R^n such that each vector u in R^n can be uniquely expressed in the form $u = v + w$ for some v in V and some w in W .

(a) Prove that 0 is the only vector in both V and W .

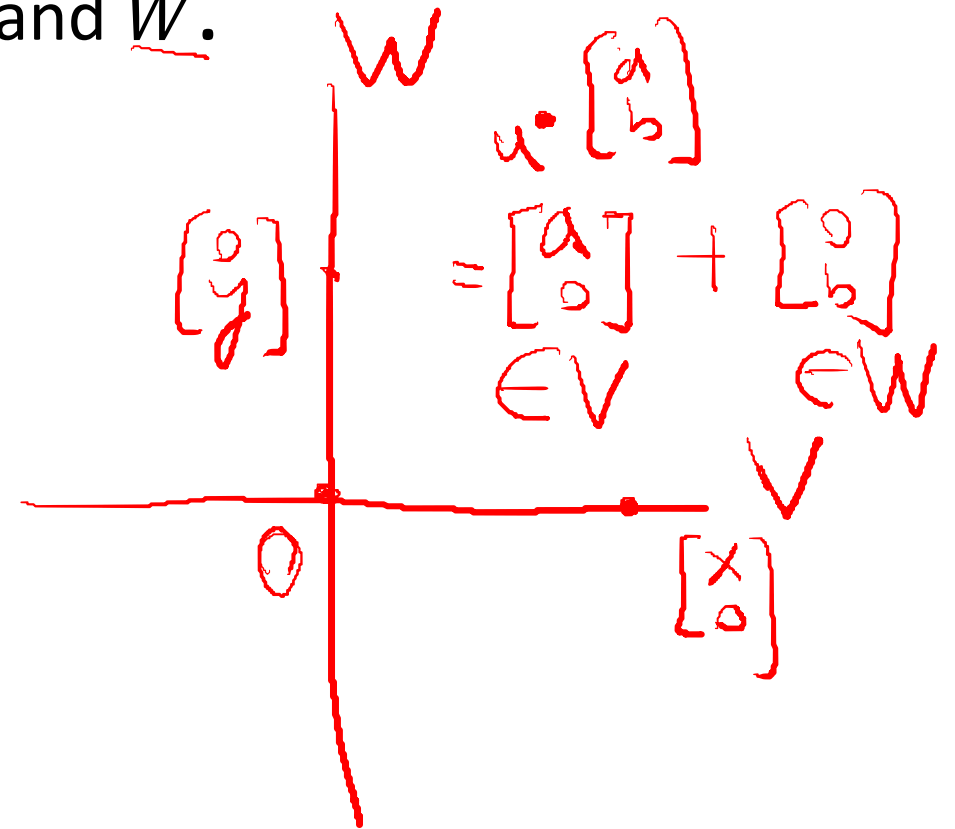
(b) Prove that $\dim V + \dim W = n$

$$\underline{u \in V}, \underline{u \in W}$$

$$u = v_1 + 0, \quad v_1 \in V, 0 \in W$$

$$u = 0 + w_1, \quad 0 \in V, w_1 \in W$$

$$\Rightarrow v_1 = 0, w_1 = 0 \Rightarrow u = 0$$



77 Let V and W be nonzero subspaces of R^n such that each vector $\mathbf{u}$ in R^n can be uniquely expressed in the form $\mathbf{u} = \mathbf{v} + \mathbf{w}$ for some $\mathbf{v}$ in V and some $\mathbf{w}$ in W .

(a) Prove that $\mathbf{0}$ is the only vector in both V and W .

(b) Prove that $\overset{=k}{\dim V} + \overset{=m}{\dim W} = \textcircled{n}$

V basis: $\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k\}$ W basis: $\{\underline{w}_1, \underline{w}_2, \dots, \underline{w}_m\}$

$\mathbb{R}^n$ basis: $\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k, \underline{w}_1, \underline{w}_2, \dots, \underline{w}_m\}$

$k + m = n$

$\mathbf{u} = \mathbf{v} + \mathbf{w} = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_k \underline{v}_k + b_1 \underline{w}_1 + b_2 \underline{w}_2 + \dots + b_m \underline{w}_m$

$\mathbb{R}^n$ is gen of $\mathbb{R}^n$

Hint: The basis of V plus the basis of W form a basis for R^n .

77 Let V and W be nonzero subspaces of R^n such that each vector u in R^n can be uniquely expressed in the form $u = v + w$ for some v in V and some w in W .

(a) Prove that 0 is the only vector in both V and W .

(b) Prove that $\dim V + \dim W = n$

$$Z = \left\{ \underbrace{v_1, v_2, \dots, v_k}_{V \text{ basis}}, \underbrace{w_1, w_2, \dots, w_m}_{W \text{ basis}} \right\}$$

$Z \subseteq R^n$ basis

Z is gen of R^n
 Z ind. $\forall v$

$x \in V$
 $x \in W$

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k + b_1 w_1 + b_2 w_2 + \dots + b_m w_m = 0$$

$$\Rightarrow c_1 v_1 + c_2 v_2 + \dots + c_k v_k = -b_1 w_1 - b_2 w_2 - \dots - b_m w_m = x = 0$$

$$\Rightarrow c_1 = c_2 = \dots = c_k = 0, -b_1 = -b_2 = \dots = -b_m = 0 \Rightarrow c_1 = \dots = c_k = b_1 = \dots = b_m = 0$$

Hint: The basis of V plus the basis of W form a basis for R^n .

78 Let V be a subspace of R^n . According to Extension Theorem, a linearly independent subset $L = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ of V is contained in a basis for V . Show that if $S = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ is any generating set for V , then the pivot columns of the matrix $[\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_m \quad \mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_k]$ form a basis for V that contains L .